

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 2 Term 2 2024

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black pen
- Calculators approved by NESA may be used
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination
- Students are permitted to refer to one double-sided A4 page of handwritten topic summaries which they prepared prior this assessment

Total marks – 49

Section I Pages 2 – 3

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Answer on the Objective Response Answer Sheet provided

Section II Pages 4 – 13

44 marks

Attempt Questions 6 -9

Allow about 52 minutes for this section

Write your answers in the space provided.

Question	1-5	6	7	8	9	Total
Total	/5	/11	/11	/11	/11	/49

This assessment task constitutes 35% of the Preliminary Higher School Certificate Course School Assessment

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

1 What is the maximum value of $y = 12 - 4x - x^2$?

A. 4

B. 8

C. 12

D. 16

2 What is $\sec^2 x - \tan^2 x - \cos^2 x$ simplified?

A. $-\tan^2 x$

B. $\sin^2 x$

C. $1 - \tan^2 x$

D. $\cot^2 x$

3 The circle $x^2 - 6x + y^2 + 2y + 6 = 0$ has:

A. centre $(3, -1)$, radius 2 units.

B. centre $(3, -1)$, radius 4 units.

C. centre $(-3, 1)$, radius 2 units.

D. centre $(-3, 1)$, radius 4 units.

- 4 How many solutions does the equation $(\sin x + 1)(2 \tan x - 3) = 0$ for $0 \leq x \leq 2\pi$ have?
- A. 2
- B. 3
- C. 4
- D. 5
- 5 Given the function $y = -\sqrt{x-1}$, the domain and the range for this function is:
- A. Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$
- B. Domain: $[1, \infty)$, Range: $(-\infty, 0]$
- C. Domain: $(1, \infty)$, Range: $(-\infty, 0)$
- D. Domain: $(0, \infty)$, Range: $(-\infty, -1)$

End of Section I

Section II

44 marks

Attempt Questions 6-10

Question 6 (11 marks)

- (a) Find the angle of inclination to the nearest degree of the line $3x - 2y - 5 = 0$ with the positive direction of the x - axis. 2

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- (b) If $\sin x = -\frac{2}{3}$, and $\tan x > 0$, find the exact value of $\cos x$. 2

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- (c) A function is given by the rule: 2

$$f(x) = \begin{cases} (x-1)^2 + 3, & x \leq 1 \\ 5x - 2, & x > 1 \end{cases}$$

What is the value of $f(f(0))$?

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Question 6 continues on page 5

Question 6 (continued)

- (d) For what values of k does the quadratic equation $3x^2 - 2x + k = 0$ have no real roots?

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- (e) Show that $\frac{\sin 150^\circ - \sin 300^\circ}{\cos 330^\circ - \cos 60^\circ} = \sqrt{3} + 2$

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End of Question 6

Question 7 (11 marks)

(a) Solve for x : $|3x + 7| = 10$

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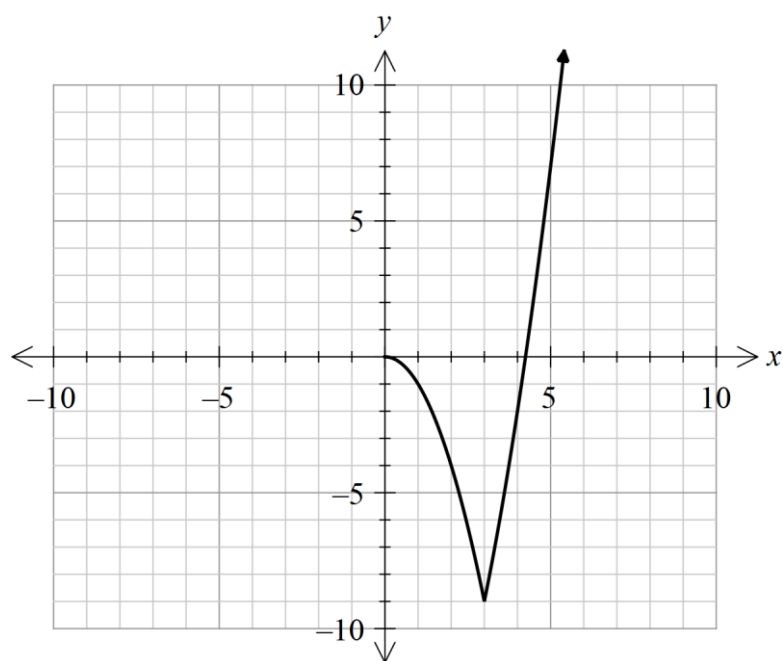
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(b) The graph of $y = f(x)$ for $x \geq 0$ is shown below.

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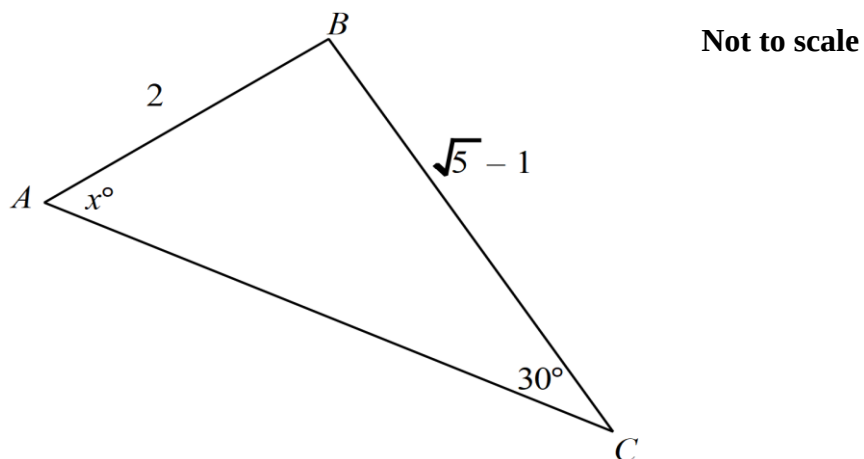
Complete the graph above for $x < 0$, given that $y = f(x)$ is an odd function for $(-\infty, \infty)$.

Question 7 continues on page 7

Question 7 (continued)

- (c) Find the value of the acute angle x in the following triangle.

2



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- (d) Determine whether the function $f(x) = \frac{x^3}{x - x^5}$ is odd, even or neither.

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Question 7 continues on page 8

(e) Find the points of intersection for $xy = 2$ and $2x + 5y = 12$.

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End of Question 7

Question 8 (11 marks)

(a) If $f(x) = 2x^2 + x - 3$ and $g(x) = x^3 + 2$, find:

(i) the degree of the polynomial $y = f(x)g(x)$.

1

.....

(ii) a simplified expression for $f(g(x))$.

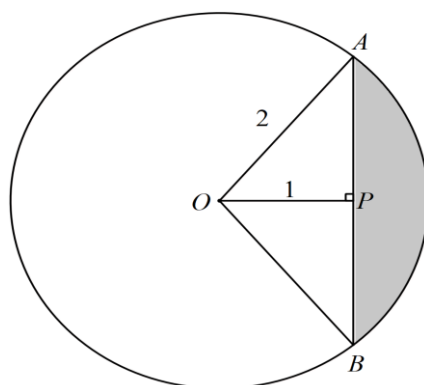
1

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(b) The diagram below shows a circle with centre O , radius 2 cm and $OP = 1$ cm.



Not to scale

(i) Find the exact area of the minor sector AOB .

2

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(ii) Hence, or otherwise, find the exact area of the shaded region.

2

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Question 8 continues on page 10

- (c) Solve $4\sin^2 \theta - 1 = 0$ for $-180^\circ \leq \theta \leq 180^\circ$. 3

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- (d) A variable y is inversely proportional to the square of x . When $x = 3$, $y = 2$. 2
Evaluate x when $y = 10$, correct to 2 decimal places, if $x > 0$.

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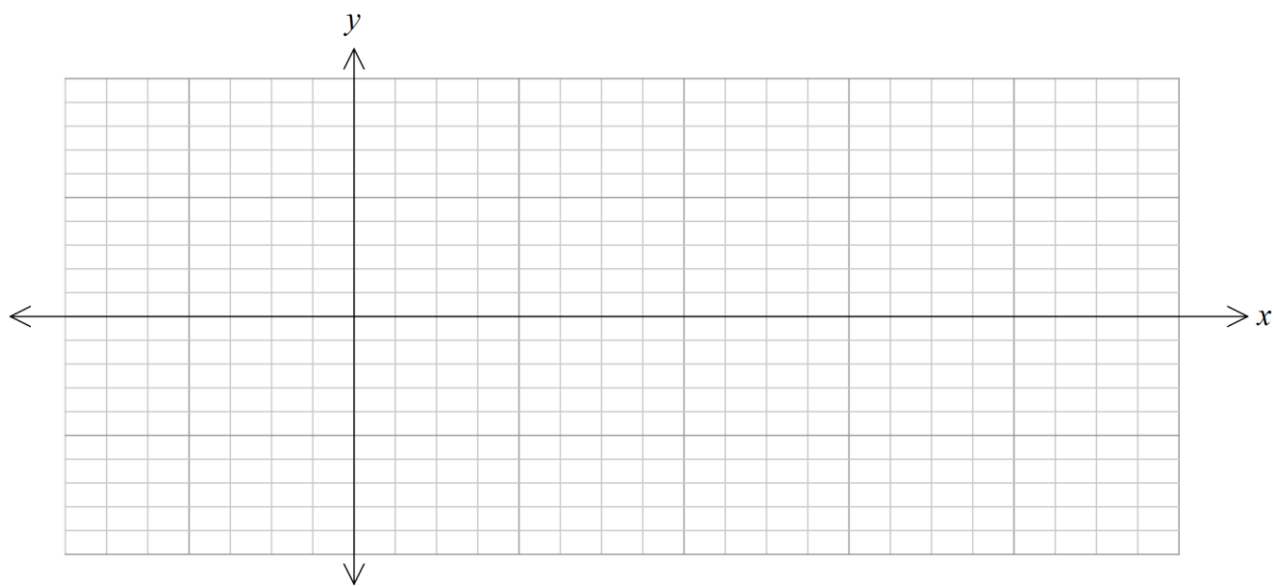
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End of Question 8

Question 9 (11 marks)

- (a) Graph $y = \cos x$ for $-\frac{\pi}{2} \leq x \leq 2\pi$ on the number plane below.

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- (b) Solve $\sin^2 x - 2 \cos x = 1$ for $-\frac{\pi}{2} \leq x \leq 2\pi$.

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Question 9 continues on page 12

- (c) Prove that $\frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$.

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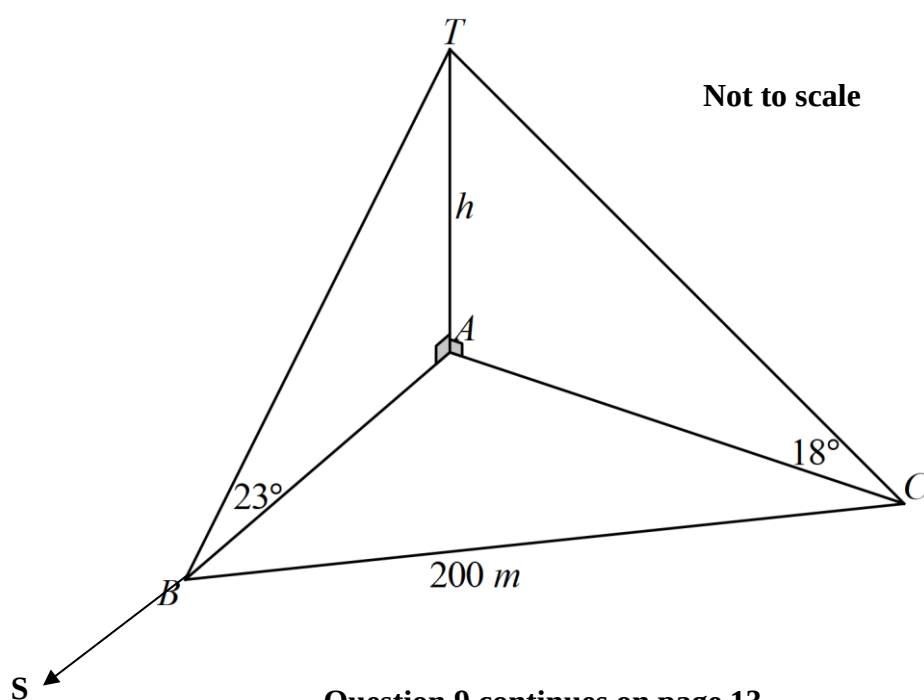
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- (d) TA is a vertical tower of height h metres.
 B and C are points at the same level as A , the foot of the tower, and $BC = 200$ m.
 The point B is due south of the tower and C is on a bearing of $120^\circ T$ from the base of the tower. The angles of elevation to the top of the tower from B and C are 23° and 18° , respectively. Find the height of the tower, correct to one decimal place.

4



Question 9 (continued)

[illegible]

End of Assessment

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HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 2 Term 2 2024

STUDENT NUMBER: _____ SOLUTIONS

General Instructions

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Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

- 1 What is the maximum value of $y = 12 - 4x - x^2$?

A. 4

B. 8

C. 12

☒ D. 16

$$\begin{aligned} x &= -\frac{b}{2a} \\ &= -\frac{(-4)}{2(-1)} \\ &= -2 \end{aligned}$$

$$\begin{aligned} \text{when } x = -2 \quad y &= 12 - 4(-2) - (-2)^2 \\ &= 12 + 8 - 4 \\ &= 16 \end{aligned}$$

58% got the correct answer.
Majority incorrect choice was C.

- 2 What is $\sec^2 x - \tan^2 x - \cos^2 x$ simplified?

A. $-\tan^2 x$

☒ B. $\sin^2 x$

C. $1 - \tan^2 x$

D. $\cot^2 x$

$$\begin{aligned} \sec^2 x - \tan^2 x - \cos^2 x \\ &= (1 + \tan^2 x) - \tan^2 x - \cos^2 x \\ &= 1 - \cos^2 x \\ &= \sin^2 x \end{aligned}$$

92% ^{gave} correct answer.
89% incorrectly chose C.

- 3 The circle $x^2 - 6x + y^2 + 2y + 6 = 0$ has:

☒ A. centre $(3, -1)$, radius 2 units.

B. centre $(3, -1)$, radius 4 units.

C. centre $(-3, 1)$, radius 2 units.

D. centre $(-3, 1)$, radius 4 units.

$$\begin{aligned} x^2 - 6x + 3^2 + y^2 + 2y + 1^2 &= -6 + 3^2 + 1^2 \\ (x - 3)^2 + (y + 1)^2 &= 4 \end{aligned}$$

$$\therefore C: (3, -1) \quad \text{Radius} = 2 \text{ units}$$

75% got the correct answer.

72% ~~of~~ incorrect answers had chosen B.

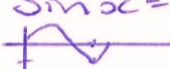
4 How many solutions does the equation $(\sin x + 1)(2 \tan x - 3) = 0$ for $0 \leq x \leq 2\pi$ have?

(A) 2

B. 3

C. 4

D. 5

$\sin x + 1 = 0$ or $2 \tan x - 3 = 0$
 $\sin x = -1$  $\tan x = \frac{3}{2}$ ~~X~~
 $x = \frac{3\pi}{2}$ $x = \tan^{-1}(\frac{3}{2})$
 Not a solution as not defined for

$\tan x$. Hence only 2 solutions.
 Poorly done. 90% of cohort chose ~~B~~ incorrectly and majority chose B forgetting $\frac{3\pi}{2}$ is undefined for $\tan \frac{3\pi}{2}$.

5 Given the function $y = -\sqrt{x-1}$, the domain and the range for this function is:

A. Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$

(B) Domain: $[1, \infty)$, Range: $(-\infty, 0]$

C. Domain: $(1, \infty)$, Range: $(-\infty, 0)$

D. Domain: $(0, \infty)$, Range: $(-\infty, -1)$

D: $x-1 \geq 0$
 $x \geq 1$

R: $y \leq 0$

Mostly done well (90% of cohort chose correct answer)

End of Section I

Section II

44 marks

Attempt Questions 6-10

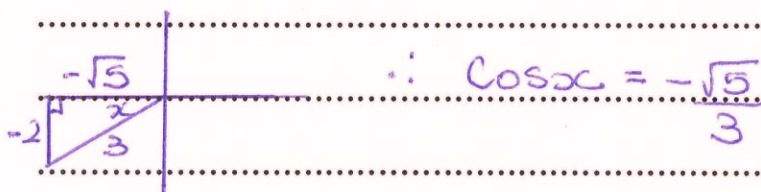
Question 6 (11 marks)

- (a) Find the angle of inclination to the nearest degree of the line $3x - 2y - 5 = 0$ with the positive direction of the x -axis. 2

mostly well done.

$$\begin{aligned}
 3x - 2y - 5 &= 0 & \tan \theta &= m \\
 -2y &= 5 - 3x & \tan \theta &= \frac{3}{2} \\
 y &= -\frac{5}{2} + \frac{3}{2}x & \theta &= \tan^{-1}\left(\frac{3}{2}\right) \\
 \therefore m &= \frac{3}{2} & & \approx 56.309^\circ \\
 & & & \approx 56^\circ \text{ (nearest degree)}
 \end{aligned}$$

- (b) If $\sin x = -\frac{2}{3}$, and $\tan x > 0$, find the exact value of $\cos x$. 2



$$\therefore \cos x = -\frac{\sqrt{5}}{3}$$

Some students didn't get the sign right. The angle is in Q3, i.e., negative cos.

- (c) A function is given by the rule: 2

$$f(x) = \begin{cases} (x-1)^2 + 3, & x \leq 1 \\ 5x - 2, & x > 1 \end{cases}$$

What is the value of $f(f(0))$?

$$\begin{aligned}
 f(0) &= (0-1)^2 + 3 \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 f(4) &= 5(4) - 2 \\
 &= 18
 \end{aligned}$$

mostly well done.

Some students didn't find the correct branch based on the x value.

Question 6 continues on page 5

Question 6 (continued)

- (d) For what values of k does the quadratic equation $3x^2 - 2x + k = 0$ have no real roots?

2

No real roots when $\Delta < 0$

$$\therefore b^2 - 4ac < 0$$

$$(-2)^2 - 4(3)(k) < 0$$

$$4 - 12k < 0$$

$$-12k < -4$$

$$\therefore k > \frac{4}{12}$$

$$k > \frac{1}{3}$$

many students understand when $\Delta < 0$, there is no real roots. However, some mistakes solving the inequality. Also, when $\Delta = 0$, there is one/double solution, so the answer shouldn't be $k > \frac{1}{3}$.

- (e) Show that $\frac{\sin 150^\circ - \sin 300^\circ}{\cos 330^\circ - \cos 60^\circ} = \sqrt{3} + 2$

3

$$= \frac{\frac{1}{2} - (-\frac{\sqrt{3}}{2})}{\frac{\sqrt{3}}{2} - \frac{1}{2}}$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)}$$

$$= \frac{\sqrt{3} + 1 + 3 + \sqrt{3}}{(\sqrt{3})^2 - (1)^2}$$

$$= \frac{4 + 2\sqrt{3}}{2}$$

$$= \frac{2(2 + \sqrt{3})}{2}$$

$$= 2 + \sqrt{3}$$

This is a show question so all steps need to be done.

Many missed writing this

And went straight to this

Other than this error most did it correctly, except for some arithmetic errors.

End of Question 6

Question 7 (11 marks)

(a) Solve for x : $|3x+7|=10$

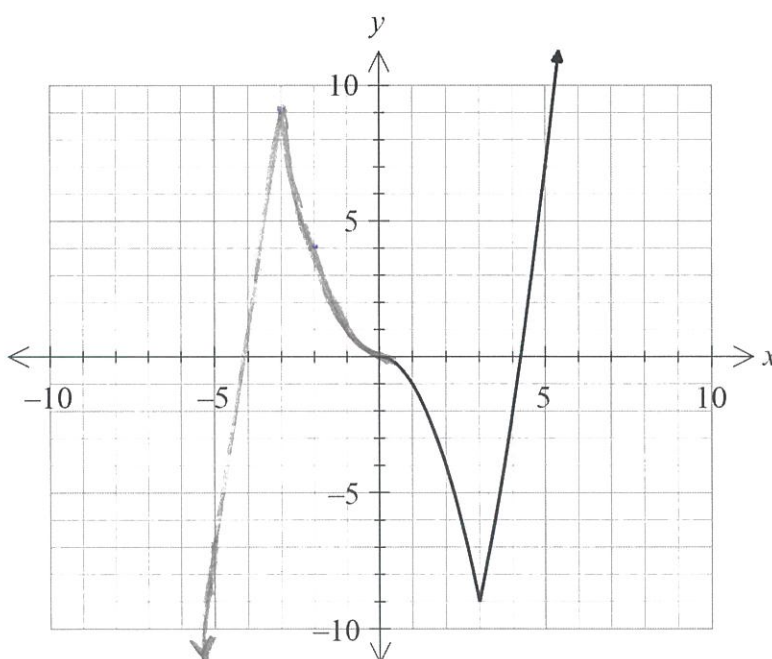
2

$$\begin{array}{l} 3x+7=10 \qquad 3x+7=-10 \\ 3x=3 \qquad 3x=-17 \\ x=1 \qquad x=-\frac{17}{3} \end{array}$$

Well done question
About 99% got this
fully correct.

(b) The graph of $y = f(x)$ for $x \geq 0$ is shown below.

2



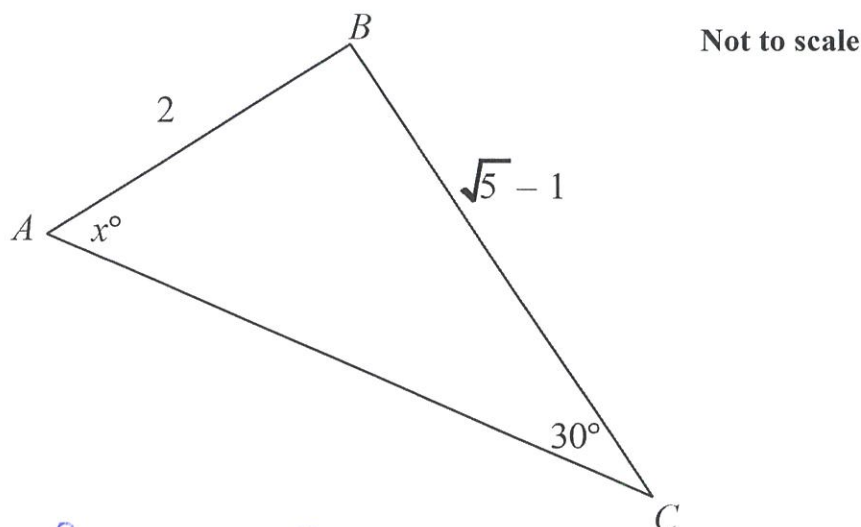
A real mixed bag here.
A lot of even functions drawn instead of odd.
Some very confused attempts.

Complete the graph above for $x < 0$, given that $y = f(x)$ is an odd function for $(-\infty, \infty)$.

Question 7 continues on page 7

- (c) Find the value of the acute angle
- x
- in the following triangle.

2



$$\frac{\sin x^\circ}{\sqrt{5}-1} = \frac{\sin 30^\circ}{2}$$

Common error NEEDS brackets

$$\sin x^\circ = \frac{1}{2} (\sqrt{5}-1)$$

$$x = 60.23^\circ$$

$$x^\circ = \sin^{-1} \left(\frac{\sqrt{5}-1}{2} \right)$$

$$= 18^\circ$$

Some students considered the ambiguous case which is good, but the question stated the acute angle. The obtuse angle $180^\circ - 18^\circ = 162^\circ$ will not work because it will exceed the angle sum of the triangle.

- (d) Determine whether the function $f(x) = \frac{x^3}{x-x^5}$ is odd, even or neither.

$$f(-x) = \frac{(-x)^3}{(-x) - (-x)^5}$$

$$= \frac{-x^3}{-x + x^5}$$

Most students showed good working for $f(-x)$

$$= \frac{-x^3}{-x + x^5}$$

$$\neq - \left(\frac{x^3}{x-x^5} \right)$$

Many students only took a single negative out as a factor.

$$= - \left[\frac{x^3}{x-x^5} \right]$$

$$\therefore f(-x) = f(x)$$

hence the function

$$= \frac{x^3}{x-x^5}$$

is even.

$$= f(x)$$

Poorly done.

Question 7 continues on page 8

(e) Find the points of intersection for $xy = 2$ and $2x + 5y = 12$.

3

$$\begin{aligned} y &= \frac{2}{x} \quad (1) & 2x + 5y &= 12 \quad (2) \\ \text{Sub (1) into (2)} & 2x + 5\left(\frac{2}{x}\right) &= 12 \\ 2x + \frac{10}{x} - 12 &= 0 \\ 2x^2 + 10 - 12x &= 0 & \text{1. for substitution to create a quadratic} \\ 2(x^2 - 6x + 5) &= 0 \\ (x-5)(x-1) &= 0 \\ \therefore x=5 \text{ or } x=1 & \text{1. for two values of } x \\ \text{When } x=5 & y = \frac{2}{5} \\ \text{When } x=1 & y = \frac{2}{1} = 2 \\ \therefore x=5, y=\frac{2}{5} \text{ or } x=1, y=2 \\ \text{The points of intersection are} \\ (5, \frac{2}{5}), (1, 2) \end{aligned}$$

End of Question 7

$$\begin{aligned} \text{OR } x &= \frac{2}{y} \\ 2 \times \frac{2}{y} + 5y &= 12 \\ 4 + 5y^2 &= 12y & \text{1.} \\ 5y^2 - 12y + 4 &= 0 \\ (5y-2)(y-2) &= 0 \\ y = \frac{2}{5} \text{ or } y &= 2 & \text{1.} \\ x = 2 \div \frac{2}{5} & & x = \frac{2}{2} \\ = 5 & & = 1 \\ (5, \frac{2}{5}), (1, 2) & \text{1.} \end{aligned}$$
$$\begin{aligned} & \text{or} \\ & \therefore \\ 2x^3 - 12x^2 + 10x &= 0 \\ 2x(x^2 - 6x + 5) &= 0 \\ 2x(x-5)(x-1) &= 0 \\ x = 0, 1, 5 \\ \text{Must give reason for} & \\ \text{discarding } x=0 & \\ \text{i.e. since } xy=2 \text{ then } x \neq 0 & \\ & y \neq 0 \end{aligned}$$

Question 8 (11 marks)

(a) If $f(x) = 2x^2 + x - 3$ and $g(x) = x^3 + 2$, find:

(i) the degree of the polynomial $y = f(x)g(x)$.

..... Degree is 5.

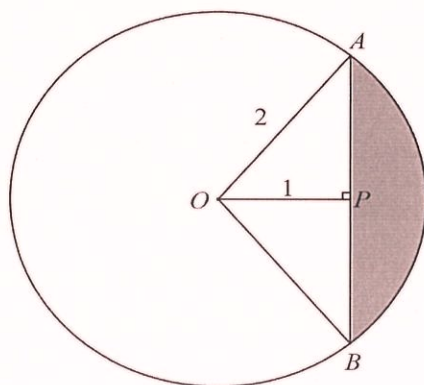
(ii) a simplified expression for $f(g(x))$.

..... $f(g(x)) = 2(x^3+2)^2 + (x^3+2) - 3$

..... $= 2x^6 + 8x^3 + 8 + x^3 - 1$

..... $= 2x^6 + 9x^3 + 7$

(b) The diagram below shows a circle with centre O , radius 2 cm and $OP = 1$ cm.



Not to scale

(i) Find the exact area of the minor sector AOB .

..... $A = \frac{1}{2} r^2 \theta$

..... $= \frac{1}{2} (2)^2 \left(120^\circ \times \frac{\pi}{180} \right)$

..... $= \frac{4\pi}{3} \text{ cm}^2$

..... OR $A = \frac{\theta}{360^\circ} \times \pi r^2$

..... $= \frac{120}{360} \times \pi (2)^2$

..... $= \frac{4\pi}{3}$

(ii) Hence, or otherwise, find the exact area of the shaded region.

..... Area of shaded region = Area of Sector - Area of triangle

..... $= \frac{4\pi}{3} - \frac{1}{2} (2)^2 \sin 120^\circ$

..... $= \frac{4\pi}{3} - 2 \times \frac{\sqrt{3}}{2}$

..... $= \frac{4\pi}{3} - 3\sqrt{3} \text{ cm}^2$

Question 8 continues on page 10

most students got this correct.

Some students did $g(f(x))$.

Some did simple arithmetic errors so didn't get final answer correct

Well answered

Some students did Area of Δ - Area of Sector

(c) Solve $4\sin^2\theta - 1 = 0$ for $-180^\circ \leq \theta \leq 180^\circ$.

3

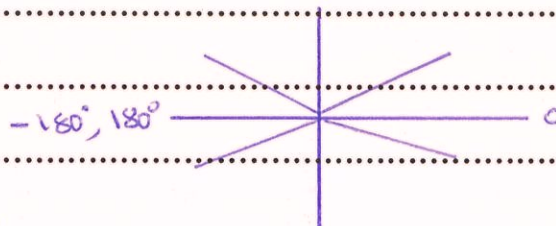
$4\sin^2\theta = 1$
 $\sin^2\theta = \frac{1}{4}$
 $\sin\theta = \pm\frac{1}{2}$

Related Acute Angle:

$\theta = \sin^{-1}\left(\frac{1}{2}\right)$
 $= 30^\circ$

$\therefore \theta = -150^\circ, \theta = -30^\circ, \theta = 30^\circ, \theta = 150^\circ$

* Completed by most quite well.



(d) A variable y is inversely proportional to the square of x . When $x = 3$, $y = 2$.

2

Evaluate x when $y = 10$, correct to 2 decimal places, if $x > 0$.

$y \propto \frac{1}{x^2}$
 $y = \frac{k}{x^2}$
 when $x = 3, y = 2$
 $2 = \frac{k}{(3)^2}$
 $\therefore k = 18$
 $\therefore y = \frac{18}{x^2}$

when $y = 10$, $10 = \frac{18}{x^2}$
 $x^2 = \frac{18}{10}$
 $x = \sqrt{\frac{18}{10}}, x > 0$
 $= 1.3416 \dots$
 ≈ 1.34 (2dp).

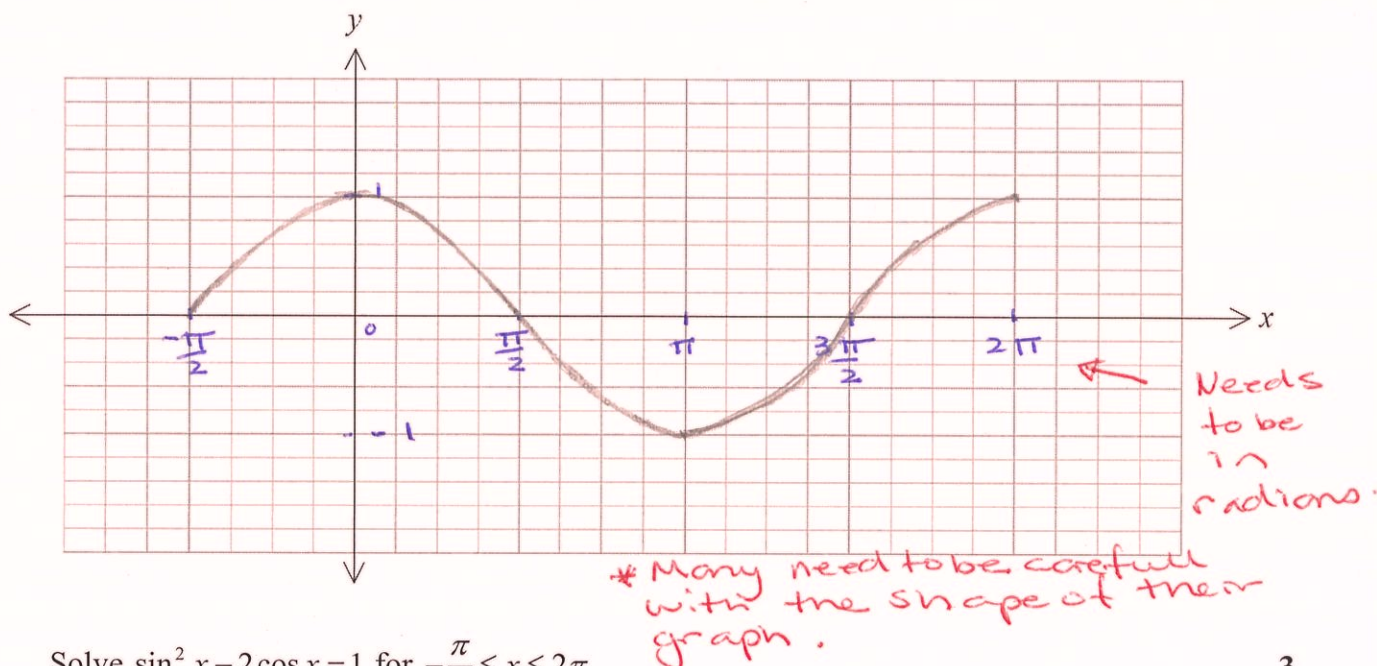
End of Question 8

Most students got
 this correct.
 Some did $y \propto \frac{1}{x}$.

Question 9 (11 marks)

- (a) Graph $y = \cos x$ for $-\frac{\pi}{2} \leq x \leq 2\pi$ on the number plane below.

2



- (b) Solve $\sin^2 x - 2\cos x = 1$ for $-\frac{\pi}{2} \leq x \leq 2\pi$.

3

$$\begin{aligned} & \dots (1 - \cos^2 x) - 2\cos x - 1 = 0 \\ & \dots 1 - \cos^2 x - 2\cos x - 1 = 0 \\ & \dots -(\cos^2 x + 2\cos x) = 0 \\ & \dots \cos x (\cos x + 2) = 0 \quad \text{Many not factorising correctly} \\ & \dots \therefore \cos x = 0 \quad \text{or} \quad \cos x + 2 = 0 \\ & \text{From graph} \quad \cos x = -2 \\ & \dots x = \cos^{-1}(0) \quad \text{No solutions as } -1 \leq \cos x \leq 1. \\ & \dots \therefore x = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2} \quad \text{Need to conclude.} \\ & \dots \quad \text{Not correct if final answer in degrees.} \end{aligned}$$

Question 9 continues on page 12

Question 9 (continued)

- (c) Prove that $\frac{\cos \theta}{1 - \sin \theta} - \tan \theta = \sec \theta$.

2

$$\begin{aligned}
 \text{LHS} &= \frac{\cos \theta}{1 - \sin \theta} - \tan \theta \\
 &= \frac{\cos \theta}{1 - \sin \theta} - \frac{\sin \theta}{\cos \theta} \\
 &= \frac{\cos^2 \theta - \sin \theta (1 - \sin \theta)}{(1 - \sin \theta) \cos \theta} \\
 &= \frac{\cos^2 \theta - \sin \theta + \sin^2 \theta}{(1 - \sin \theta) \cos \theta} \\
 &= \frac{1 - \sin \theta}{(1 - \sin \theta) \cos \theta} \\
 &= \frac{1}{\cos \theta} \\
 &= \sec \theta \\
 &= \text{RHS}
 \end{aligned}$$

Mostly done well.

Students had problem with this proof had poor technique,

leaving denominator off fraction and ~~often~~ were

unable to arrive to correct

required answer.

Some students had problems

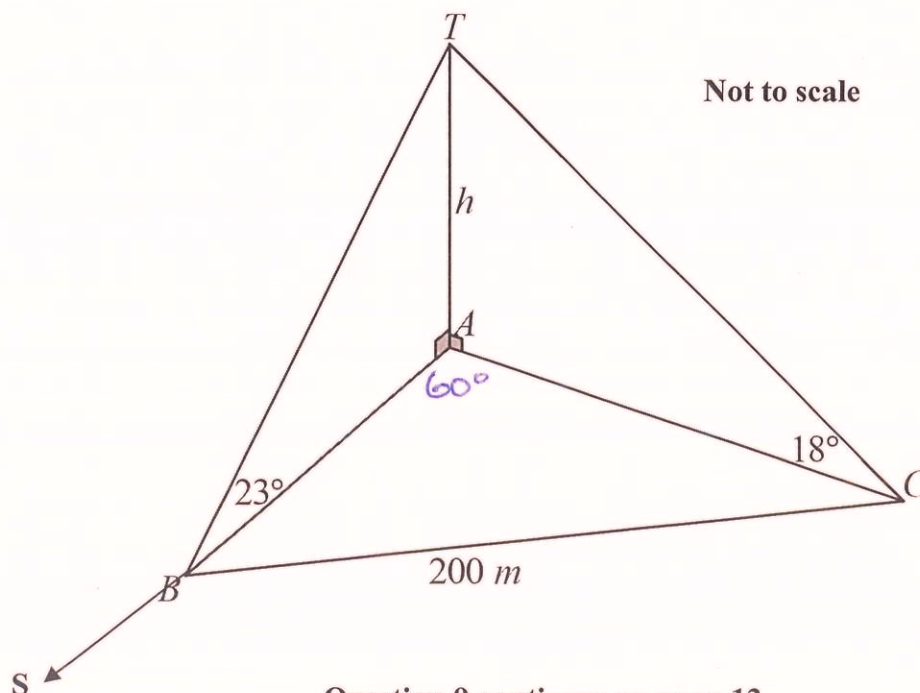
with trigonometric identities.

- (d) TA is a vertical tower of height h metres.

4

B and C are points at the same level as A , the foot of the tower, and $BC = 200$ m.

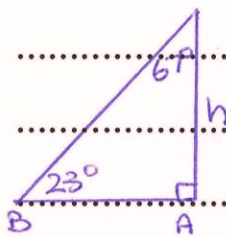
The point B is due south of the tower and C is on a bearing of $120^\circ T$ from the base of the tower. The angles of elevation to the top of the tower from B and C are 23° and 18° , respectively. Find the height of the tower, correct to one decimal place.



Question 9 continues on page 13

Question 9 (continued)

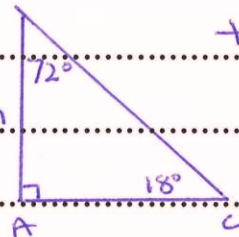
In $\triangle TAB$:



$$\tan 67^\circ = \frac{BA}{h}$$

$$\therefore BA = h \tan 67^\circ$$

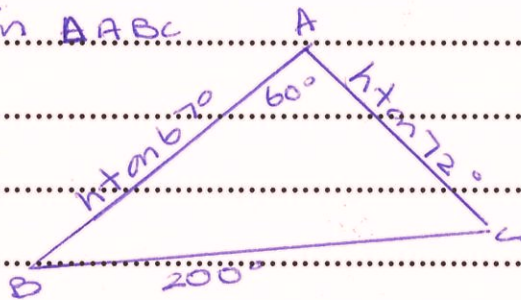
In $\triangle TAC$:



$$\tan 72^\circ = \frac{AC}{h}$$

$$\therefore AC = h \tan 72^\circ$$

In $\triangle ABC$:



$$BC^2 = AB^2 + AC^2 - 2 \times AB \times AC \times \cos \angle BAC$$

$$200^2 = (h \tan 67^\circ)^2 + (h \tan 72^\circ)^2 - 2 \times h \tan 67^\circ \times h \tan 72^\circ \times \cos 60^\circ$$

$$200^2 = h^2 (\tan^2 67^\circ + \tan^2 72^\circ - \tan 67^\circ \tan 72^\circ)$$

$$\therefore h^2 = \frac{200^2}{\tan^2 67^\circ + \tan^2 72^\circ - \tan 67^\circ \tan 72^\circ}$$

$$= \frac{200^2}{5.1461939803}$$

$$\therefore h = \sqrt{5146.1939803}, h > 0$$

$$= 71.74217 \dots$$

$$\approx 71.7 \text{ m (1dp)}$$

Mostly did get ad presented work successfully.

Students with incorrect answer were due to calculator error or had problem in solving for h when using cosine rule $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$, especially with $\cos 23^\circ$ and $\cos 18^\circ$.

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